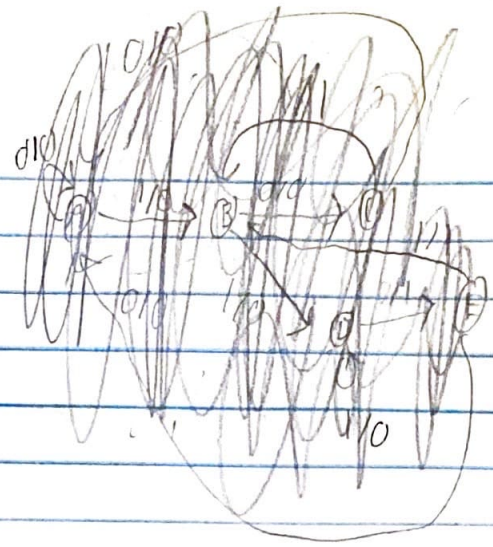
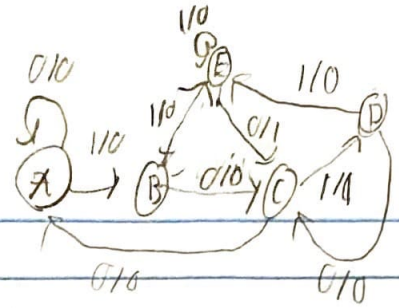
1.





Dylan Trumby

P state	Next W=0	Next W=1	Z
A	A	B	0
B	C	E	0
C	A	D	1 if from E
D	C	E	1 if from C
E	C	E	0

2.

W	$Q_2Q_1Q_0$	next states $Q_2Q_1Q_0$	out	D_{Q_2}	J_{Q_1}	K_{Q_1}	J_{Q_0}
0	000	101	1	1	0d		1
0	010	111	0	1	d0		1
0	011	000	1	0	d1		1
0	101	011	0	0	1d		0
0	111	111	1	1	d0		0
1	000	010	0	0	1d		0
1	010	011	1	0	d0		1
1	011	010	0	0	d0		1
1	101	111	1	1	1d		0
1	111	000	1	0	d1		1

X	Q(t+1)
0	Q(t)
1	$\overline{Q(t)}$

Q(t)	Q(t+1)	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

out

	Q_1Q_0 00	01	11	10
00	1	d	1	0
01	d	0	1	d
11	d	1	1	d
10	0	d	0	1

$W\overline{Q_2}$

$WQ_2 + \overline{W}\overline{Q_1}\overline{Q_0} +$

$\overline{W}Q_1Q_0 + WQ_1\overline{Q_0}$

$\boxed{W(Q_2 + Q_1\overline{Q_0}) + \overline{W}(\overline{Q_1 \oplus Q_2})}$ = out

D_{Q_2}

	Q_1Q_0 00	01	11	10
00	0	d	d	d
01	d	1	d	d
11	d	1	d	d
10	1	d	d	d

J_{Q_1}

	Q_1Q_0 00	01	11	10
00	d	d	1	0
01	d	d	0	d
11	d	d	1	d
10	d	d	0	0

K_{Q_1}

$J = Q_2 + W$

$K = WQ_2 + \overline{W}\overline{Q_2}Q_0$

out =

$D = \overline{W}\overline{Q_0} + \overline{W}Q_2Q_1 + WQ_2\overline{Q_1}$

$\boxed{D = \overline{W}\overline{Q_0} + Q_2(W \oplus Q_1)}$

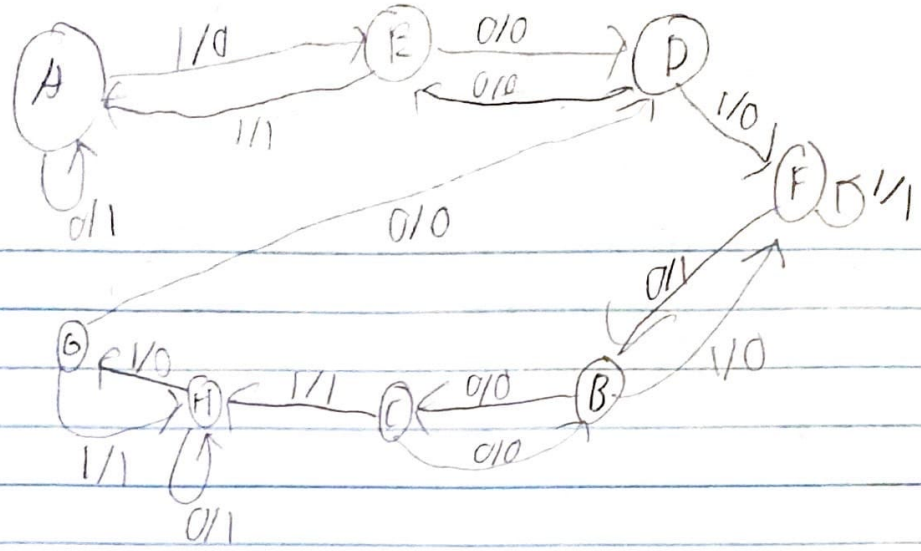
J_{Q_0}

	Q_1Q_0 00	01	11	10
00	1	d	1	1
01	d	0	0	d
11	d	0	1	d
10	0	d	1	1

$\boxed{T = \overline{W}Q_2 + WQ_1}$

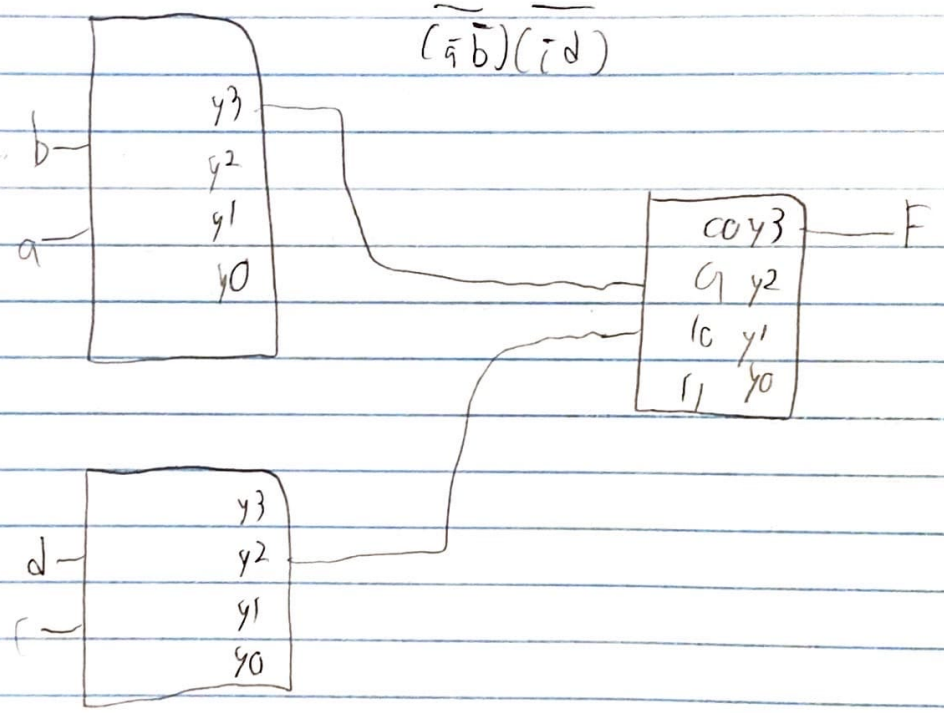
x/z
↑

3.

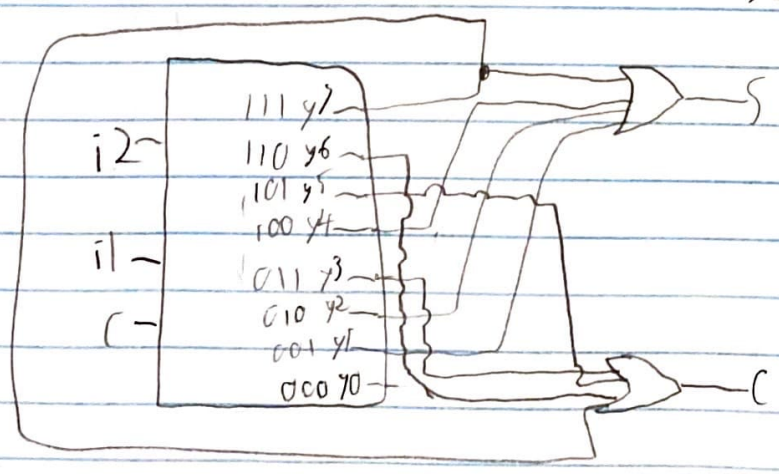


4.

$(\overline{\bar{a}\bar{b}})(\overline{\bar{c}d})$



5.

$S = y7+y4+y2+y1$

$C = y7+y6+y5+y3$